Homo ergaster was closely related to *Homo erectus*, the first humans to spread out of tropical Africa into Europe and Asia as part of a general radiation of mammals and their predators some 1.8 mya. *Homo erectus* was a skilled hunter and a brilliant opportunist, quick to take advantage of different environments—a key factor in the success of the human species.

These early humans soon settled in South and Southeast Asia, reaching Dmanisi in Georgia by 1.7 mya (see pp.24–25). They were well established in Western Europe by at least 800,000 years ago. Warmer conditions than today may have attracted *Homo heidelbergensis* to Northern Europe by 400,000 years ago. At about the same time, small bands of early humans were using long-shafted, aerodynamic wooden spears to hunt wild horses and larger game at Schöningen, Germany, and at Boxgrove in southern England.

The remarkable finds at Schöningen are the earliest preserved wooden tools yet discovered. *Homo heidelbergensis* lived in small, mobile groups. Each group probably returned to the same locations to hunt and forage at different times of the year. However, their communication and reasoning abilities were limited (see pp.20–21), which affected their ability to adapt and may be one reason why they do not appear to have settled in intensely cold environments or reached the Americas and Australia.

Adapting to different environments

By 500,000, early humans had adapted successfully to a wide variety of tropical and temperate environments, moving as far north as China, where numerous fragments of an evolving *Homo erectus* have come to light in Zhoukoudian Cave, near Beijing. The ability to use fire (see p.16) was crucial in making settlement possible in cold locations

HOW WE KNOW

THEY WALKED ON TWO FEET

About 3.75 mya, a volcanic eruption left a layer of ash at Laetoli, Tanzania that preserved the footprints of *Australopithecus afarensis* ("Lucy"). They were identified as those of a young adult who walked on two feet with a rolling gait, slower than that of modern humans. This bipedal posture—an important human anatomical feature that appeared before 4 mya—allowed our ancestors to live away from forests in open terrain.

Human family tree

New discoveries of fossils that add to our knowledge of human evolution are being made all the time. The size and shape of the skulls help us to understand the abilities of our ancestors. Brain size is measured in cubic centimeters (cc), with an average modern human brain measuring 1,400 cc.

6.2–5.8 mya

Sites Africa

Brain size Unknown

Orrorin tugenensis is known to us through finds of large canine teeth. Little is known about the species, except that it may have been bipedal.

5.8–5.2 mya

Sites Africa

Brain size Unknown

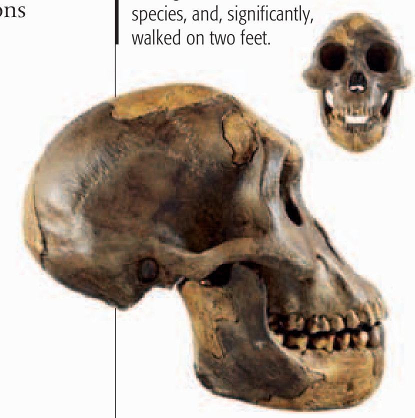
Ardipithecus kadabba was one of the earliest species to be placed on the human tree. Like *Orrorin tugenensis*, this species had primitive canine teeth.

3–2.4 mya

Site Africa

Brain size 375–500 cc

Australopithecus afarensis
Known as "Lucy," this early hominin was relatively short at 3 ft 3 in (1 m) in height, had shorter limbs than later species, and, significantly, walked on two feet.

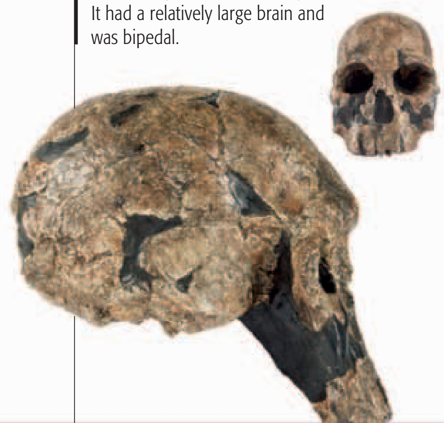


2.5–1.8 mya

Sites Africa

Brain size 750 cc

Homo rudolfensis, a contemporary of *Homo habilis*, has been the subject of much debate concerning its age and relationship to the hominin species. It had a relatively large brain and was bipedal.



Homo rudolfensis

Ardipithecus kadabba

Orrorin tugenensis

Australopithecus afarensis

7 MYA | 6 MYA | 5 MYA | 4 MYA | 3 MYA | 2.5 MYA | 2 MYA

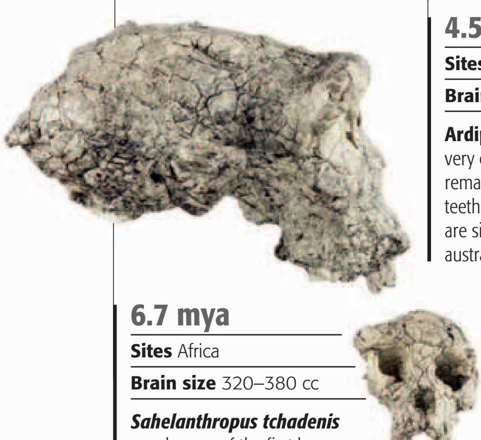
Sahelanthropus tchadensis

Ardipithecus ramidus

Australopithecus africanus

Australopithecus anamensis

Homo habilis



6.7 mya

Sites Africa

Brain size 320–380 cc

Sahelanthropus tchadenis
may be one of the first humans or may be more closely related to apes, as it shows a mixture of human and ape characteristics. Only the fragments of a skull have been found.

4.5–4.3 mya

Sites Africa

Brain size Unknown

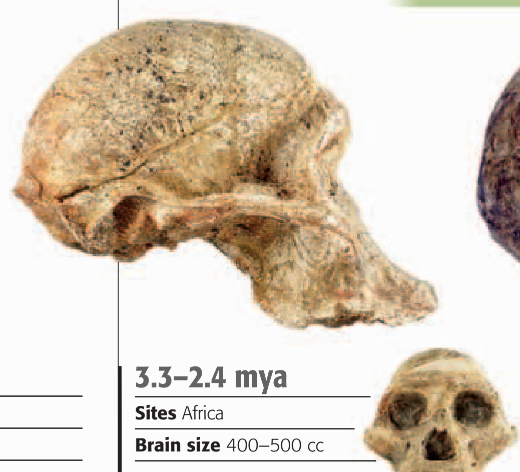
Ardipithecus ramidus is a very early hominin. Fragmentary remains include large canine teeth found in Ethiopia, which are similar to those of the australopithecines.

4.3–4 mya

Sites Africa

Brain size Unknown

Australopithecus anamensis
is little known as few remains have been found. The jawbone from Kenya resembles that of a chimpanzee, while the teeth are closer to human teeth.



3.3–2.4 mya

Sites Africa

Brain size 400–500 cc

Australopithecus africanus
was a slenderly built species. Its facial features appear to have been more human than earlier australopithecines. It had longer legs and shorter arms than modern humans.

2.5–1.8 mya

Sites Africa

Brain size 590–650 cc

Homo habilis had relatively long arms, marking it out from later humans. The species may descend from the australopithecines.